

# ORF 309 Homework 6 - Corrections

## Corrected Solutions

### Problem 4b: Conditional Distribution of $N_t$ given $N_s = n$

#### Your Error

For the case when  $t < s$ , you wrote:

$$P(N_t = k \mid N_s = n) = \binom{n}{k} \left(\frac{k}{s}\right)^k \left(1 - \frac{k}{s}\right)^{n-k}$$

This is incorrect. You used  $k/s$  instead of  $t/s$  in the binomial probability.

#### Correct Answer

For  $t < s$ :

Given  $N_s = n$ , the  $n$  arrivals are uniformly distributed on  $[0, s]$ . Each arrival has probability  $\frac{t}{s}$  of occurring before time  $t$ . Therefore,  $N_t \mid N_s = n$  follows a binomial distribution with parameters  $n$  and  $p = \frac{t}{s}$ .

$$P(N_t = k \mid N_s = n) = \binom{n}{k} \left(\frac{t}{s}\right)^k \left(1 - \frac{t}{s}\right)^{n-k}, \quad k = 0, 1, \dots, n$$

#### Complete Answer for Part (b)

The conditional distribution of  $N_t$  given  $N_s = n$  depends on whether  $t \geq s$  or  $t < s$ :

- **For  $t \geq s$ :**  $N_t \mid N_s = n \sim n + \text{Poisson}(\lambda(t - s))$

$$P(N_t = k \mid N_s = n) = \frac{[\lambda(t - s)]^{k-n} e^{-\lambda(t-s)}}{(k - n)!}, \quad k = n, n + 1, n + 2, \dots$$

- **For  $t < s$ :**  $N_t \mid N_s = n \sim \text{Binomial}\left(n, \frac{t}{s}\right)$

$$P(N_t = k \mid N_s = n) = \binom{n}{k} \left(\frac{t}{s}\right)^k \left(1 - \frac{t}{s}\right)^{n-k}, \quad k = 0, 1, \dots, n$$

## Problem 5a: Joint Distribution and Independence of $R_n$ and $G_n$

### Your Error

You stated: “They are independent”

**This is incorrect.**  $R_n$  and  $G_n$  are **NOT** independent.

### Correct Answer

Let  $R_n$  be the number of times the monkey pressed the red button and  $G_n$  be the number of times the green button was pressed after  $n$  total button presses.

### Joint Distribution

$$P(R_n = r, G_n = g) = \begin{cases} \binom{n}{r} \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^g & \text{if } r + g = n, r, g \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Note: When  $r + g = n$ , we have  $g = n - r$ , so this can also be written as:

$$P(R_n = r, G_n = n - r) = \binom{n}{r} \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^{n-r}$$

### Independence

$R_n$  and  $G_n$  are **NOT** independent.

**Proof:**

There is a deterministic constraint:  $R_n + G_n = n$  always holds.

This means if you know  $R_n = r$ , you automatically know  $G_n = n - r$ . They are perfectly dependent.

More formally, for independence we would need:

$$P(R_n = r, G_n = g) = P(R_n = r) \cdot P(G_n = g) \quad \text{for all } r, g$$

But this fails. For example:

- The left side is **zero** when  $r + g \neq n$
- However,  $P(R_n = r) > 0$  and  $P(G_n = g) > 0$  for many values where  $r + g \neq n$
- So the right side is positive while the left side is zero
- Therefore, the equality fails and they are not independent

## Marginal Distributions

For completeness:

$$R_n \sim \text{Binomial}\left(n, \frac{1}{3}\right) : \quad P(R_n = r) = \binom{n}{r} \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^{n-r}$$

$$G_n \sim \text{Binomial}\left(n, \frac{2}{3}\right) : \quad P(G_n = g) = \binom{n}{g} \left(\frac{2}{3}\right)^g \left(\frac{1}{3}\right)^{n-g}$$

## Problem 5b: Why ARE $R_N$ and $G_N$ Independent?

This is a beautiful contrast to part (a). When  $N$  is random (Poisson),  $R_N$  and  $G_N$  **ARE independent**.

### Intuition

The key difference:

- In part (a), the **total number of button presses  $n$  is fixed**, creating the constraint  $R_n + G_n = n$
- In part (b), the **total number  $N$  is random and independent** of the colors chosen
- By the thinning property of Poisson processes,  $R_N$  and  $G_N$  become independent Poisson random variables

### Result

$$R_N \sim \text{Poisson}\left(\frac{a}{3}\right)$$
$$G_N \sim \text{Poisson}\left(\frac{2a}{3}\right)$$

And they are **independent**, so:

$$P(R_N = r, G_N = g) = P(R_N = r) \cdot P(G_N = g) = \frac{(a/3)^r e^{-a/3}}{r!} \cdot \frac{(2a/3)^g e^{-2a/3}}{g!}$$

This can be simplified to:

$$P(R_N = r, G_N = g) = \frac{a^r}{3^r r!} \cdot \frac{(2a)^g}{3^g g!} \cdot e^{-a}$$

### Key Takeaway

- **Fixed total  $n$ :**  $R_n$  and  $G_n$  are NOT independent (constraint:  $R_n + G_n = n$ )
- **Random total  $N$ :**  $R_N$  and  $G_N$  ARE independent (Poisson thinning)